# Network-Aware Predictive Digital Twin Framework for Resilient and Self-Reconfigurable Cyber-Physical Production Systems

**Abstract.** Sensor faults and delayed observations can make supervisory decisions inconsistent with the evolving state of a production cell. This paper presents a network-aware predictive digital twin framework and its simulation-stage implementation for a conveyor-based sorting and pick-and-place system. The architecture separates a MuJoCo plant, timestamped sensor observations, a communication model, state estimation, sequence control, and independent evaluation. A supervisory predictor propagates position uncertainty and observation age through 54 candidate one-dimensional stopping trajectories over a 0.5 s horizon. It selects whether to continue for one 20 ms control interval or request a latched hold; incomplete predictions are rejected when their computation budget is exceeded. Four end-to-end acceptance cases, comprising seven simulation runs, exercise nominal operation, a photoelectric sensor stuck low, insufficient motion feedback, and complete telemetry loss. In the matched sensor-fault case, predictive supervision requests a hold 40 ms earlier than the sequence-control baseline. Both nominal runs complete placement in 15.46 s of simulated time, providing no observed cycle-time improvement in that comparison. These results demonstrate an executable, traceable basis for investigating software-level supervisory adaptation. They do not establish fault-diagnosis accuracy, physical safety, or autonomous recovery. Physical calibration, live programmable logic controller integration, and recovery-policy validation remain necessary before deployment as a production digital twin.

**Keywords:** Digital twin · Networked control · State estimation · Predictive supervision

## 1 Introduction

Digital twins offer a means of connecting the state of a production system to an executable representation that can support monitoring and operational decisions. The distinction between a digital model, a digital shadow, and a digital twin depends on how the physical and digital systems exchange information [2]. For a production cell, a geometric representation alone therefore does not establish a working digital twin. The representation must also support explicit assumptions about observations, state evolution, and the actions that may be executed. In this

work, the term digital twin framework denotes an architecture developed toward that integration; the evaluated implementation remains a simulation testbed without a live connection to the physical controller.

Communication is part of this integration problem. Networked control must account for sampling, variable delay, packet loss, and potentially reordered observations rather than assuming ideal channels [1]. Information freshness provides a related perspective: a received update describes a state generated at an earlier time, and its usefulness depends on the age of that information [4]. In a handling cell, the distinction has a concrete consequence. A workpiece may continue moving between a sensor event and the supervisory decision based on that event. A missing photoelectric transition may also result from a sensor fault, an observation that has not yet arrived, or motion that differs from the assumed model. These alternatives should not be treated as interchangeable evidence of a diagnosed fault.

The application considered here comprises two conveyors, a pneumatic transfer pusher, an XY gantry with a vertical actuator, and a vacuum gripper. The production sequence transfers one workpiece between conveyors and subsequently places it in a selected bin. The immediate research question is deliberately bounded: can a supervisor use timestamped observations and a finite set of predicted stopping trajectories to decide whether one additional transport interval remains admissible when sensing or communication is disturbed? This formulation addresses the operational decision before attempting broader claims about fault isolation, throughput optimization, or autonomous recovery. In particular, a supervisory method should not rely on the true workpiece position available inside a simulator, because that information is unavailable to a controller with only sparse physical observations.

To investigate this question, the framework separates plant simulation, sensing, communication, estimation, sequence control, prediction, and evaluation. MuJoCo provides the multibody and contact simulation environment; its design supports model-based control and contact dynamics [3]. In the implemented testbed, photoelectric events anchor a one-workpiece position estimator, while timestamped messages pass through a configurable delay, jitter, and packet-loss model. Simulator ground truth is reserved for generating observations and evaluating performance. The controller and supervisor operate on received observations and estimated state. Evaluation associates estimates with the corresponding source-time state, avoiding a comparison between a delayed estimate and a newer plant position.

The predictive supervisor uses a reduced one-dimensional stopping model rather than repeated full-contact MuJoCo simulations. Candidate trajectories combine position offsets, velocity offsets, reaction delays, and two supervisory actions. The continue action advances transport for one control interval before an assumed stop; the hold action requests an immediate stop subject to the modeled reaction delay and coast allowance. The supervisor compares the worst predicted stopping margin with the modeled support limit and executes only the first decision before replanning. A missed computation budget produces a hold

request rather than acceptance of a partially evaluated candidate set. The uncertainty allowance is an engineering bound awaiting calibration, not a validated probability interval or a physical safety certificate.

The present contribution has three parts. First, it provides an executable architecture with an explicit separation of controller information from simulator state reserved for evaluation. Second, it implements a supervisory rule constrained by a computation budget, incorporating observation age and bounded motion uncertainty into selection between continuation and a hold. Third, it provides paired experiments with recorded configurations, in which baseline and predictive supervision use the same requested operating conditions and network seed. These contributions support reproducible evaluation of supervisory adaptation without changing the mechanical configuration. At the current stage, adaptation is limited to continuation or a hold; automatic sensor substitution, fault recovery, and online optimization of conveyor speed remain outside the demonstrated capability.

Four end-to-end acceptance cases, comprising seven simulation runs, establish the scope of the initial evidence. Nominal baseline and predictive runs both complete placement in 15.46 s of simulated time. With the tested photoelectric sensor held low, predictive supervision requests a hold at 6.72 s, compared with 6.76 s for the sequence-control timeout. Separate cases exercise missing motion confirmations and complete telemetry loss. These observations demonstrate the implemented decision path under specified fixtures; they do not estimate fault-detection accuracy or prove a general improvement in production resilience. The experiments use an uncalibrated baseline geometry and a shared simulation clock. Subsequent dimensional measurements and CAD revisions have not yet been propagated into this evaluated plant. Physical validation and recovery-policy experiments are therefore required before claiming a deployed, self-reconfiguring production system.

## References

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